

TASK 1: Linux Environment Setup & Exploration

Objective

To understand basic Linux installation, file system navigation, file operations, permissions, and system monitoring using Ubuntu Linux.

1 Linux Installation & Login

Option A: Ubuntu on VirtualBox / VMware

- Download **Ubuntu 22.04 LTS**
- Create a new Virtual Machine

VM Configuration:

-  **RAM: 2 GB**
-  **CPU: 2 Cores**

Complete the Ubuntu installation, create a user account, and successfully log in.

2 Terminal Access & Basic Commands

Open Terminal and verify login details:

- Check logged-in user:
whoami
- Check hostname:
hostname
- Check current directory:

Pwd

3 File & Directory Exploration

List Files

ls

ls -l

ls -a

Navigate Directories

cd /

cd /home

cd ~

Root vs Home Directory

Path	Description
/	Root directory (top of filesystem)
/home/u sername	User workspace

4 Create Directories & Files

Create Directory

```
mkdir linux_task
```

```
cd linux_task
```

Create Files

```
touch file1.txt file2.txt
```

Editors Used

- **Nano** (Beginner friendly)
- **Vim** (Advanced editor)

Verify files:

```
ls
```

5 File Viewing & Editing

View File Content

```
cat file2.txt
```

```
less file2.txt
```

Edit Using Nano

```
nano file2.txt
```

- Save: **CTRL + O**
- Exit: **CTRL + X**

Edit Using Vim (Basic)

```
vim file2.txt
```

- Insert mode: **i**
- Save & Exit: **:wq**

6 File & Directory Deletion

- Remove a file safely:

```
rm file1.txt
```

- Remove empty directory:

```
rmdir empty_folder
```

- Remove directory with files:

```
rm -r linux_task
```

7 File Permissions & Ownership

Check Permissions

```
ls -l
```

Example:

```
-rw-r--r-- 1 user user 12 file.txt
```

Permission Meaning

Symbol	Meaning
r	Read
w	Write
x	Execute

Change Permissions

chmod 755 file.txt

chmod u+x file.txt

Change Ownership (Root Only)

sudo chown root:root file.txt

8 System Monitoring Basics

CPU & Process Usage

top

Better UI (Optional)

sudo apt install htop

htop

Disk Usage

df -h

du -sh *

Memory Usage

free -m

9 Linux Directory Structure

Directory	Purpose
/	Root filesystem
/home	User files
/etc	Configuration files
/var	Logs & variable data
/bin	Essential binaries
/usr	User programs

♦ **How do you check disk usage?**

`df -h`

`du -sh foldername`

♦ **What happens when permission is denied?**

The user lacks required permissions to access the file or directory.

Solutions:

- `chmod` – change permissions
- `sudo` – run as root
- `chown` – change ownership